MA1522 LINEAR ALGEBRA FOR COMPUTING
SEMESTER 1, AY2026/2027

CHI-HENG LO
ch_lo@nus.edu.sg
S17 # 07-12

This course is targeted at students from the School of Computing. It introduces the basic concepts of linear algebra that are routinely applied in computer science, in the context of Euclidean spaces. The course focuses on the theory behind each algorithm, equipping students to compute with vectors and matrices by hand and to use modern computational and AI tools effectively with genuine understanding.

The teaching mode for this course is **Blended Learning**. This means that students will view recorded lecture videos at their own convenience, and then use 'Lecture' hours to consolidate their understanding by attempting questions posed by their lecturers on the spot. This differs from the traditional lecture-cum-tutorial format, where students are typically expected to attempt tutorial questions as homework before attending tutorials. In the Blended Learning format, the homework instead consists of viewing the recorded lecture videos before attending 'Lecture'.

Learning Outcomes. By the end of this course, students should be able to:

- solve systems of linear equations and carry out matrix computations, both by hand and using computational software (e.g. MATLAB);
- work confidently with vectors, matrices, determinants, and Euclidean spaces;
- reason about subspaces, linear independence, bases, dimension, and the rank of a matrix;
- compute inner products, eigenvalues and eigenvectors, and diagonalise matrices;
- analyse linear transformations between Euclidean spaces and apply these tools to problems in computing.

Reference. The course follows the lecture slides and recorded videos provided on Canvas; no separate textbook is required.

Schedule for Group L1.
Lectures: Tuesdays 10:00–12:00 and Fridays 8:00–10:00. See the lesson schedule at the end of this document for the full weekly breakdown of sections covered.

Grades. The final grade will be generated from the following components:

- *Participation:* 20%.
- *Online Quizzes:* 10%.
- *Final Exam:* 70%.

Participation. Students are expected to submit solutions to the problems discussed during each lecture of their respective lecture groups, to claim credit for participation.

- Each eligible submission, submitted by the end of each lecture and graded for participation rather than correctness, will be awarded 1%, capped at 20% across all submissions.
- As students can only submit to solutions according to which lecture group they belong to, they will not be able to make up for missed submissions by submitting to the other group.

- Appeals for missed submissions will **NOT** be handled on an ad-hoc basis, but together during the Reading Week.
- Appeals for failed submissions due to Canvas system error will **NOT** be granted. As such, always submit early and ensure your submission is successful.

Online Quizzes. There will be five online quizzes, each worth 2.5%, with the best four scores counted toward the final grade. Each quiz opens at 0:00 and is due at 23:59 on the following day:

- Quiz 1 — available 29 Aug, due 30 Aug 2026.
- Quiz 2 — available 12 Sep, due 13 Sep 2026.
- Quiz 3 — available 3 Oct, due 4 Oct 2026.
- Quiz 4 — available 17 Oct, due 18 Oct 2026.
- Quiz 5 — available 31 Oct, due 1 Nov 2026.

Final Exam. The final exam is closed book (one helpsheet permitted) and will take place on 24 November 2026, 5:00 pm. It will cover material from the entire course. More details will be announced near the end of the semester.

Important Dates.

Date	*Event*
30 Aug 2026	Quiz 1 due (23:59)
13 Sep 2026	Quiz 2 due (23:59)
4 Oct 2026	Quiz 3 due (23:59)
18 Oct 2026	Quiz 4 due (23:59)
1 Nov 2026	Quiz 5 due (23:59)
24 Nov 2026	*Final Exam* (5 pm)

Lesson Schedule.
This is a tentative schedule for the problems I will give in each lesson; any changes will be announced in class. You are expected to finish learning the materials on Canvas a lesson ahead of the dates shown below.

Week	*Lesson*	*Date*	*Day*	*Sections covered*
1	1	11 Aug 2026	Tue	Sections 1.1–1.3
	2	14 Aug 2026	Fri	Sections 1.4 & 1.5
2	3	18 Aug 2026	Tue	Sections 2.1–2.3
	4	21 Aug 2026	Fri	Sections 2.4–2.6
3	5	25 Aug 2026	Tue	Sections 2.7–2.9
	6	28 Aug 2026	Fri	Sections 2.8–2.10
4	7	1 Sep 2026	Tue	Sections 3.1–3.3
	8	4 Sep 2026	Fri	Sections 3.3 & 3.4
5	9	8 Sep 2026	Tue	Sections 3.4–3.5
	10	11 Sep 2026	Fri	Sections 3.6 & 3.7
6	11	15 Sep 2026	Tue	Sections 3.7 & 3.8
	12	18 Sep 2026	Fri	Sections 4.1 & 4.2
Recess / Reading Week — Sat 19 Sep – Sun 27 Sep 2026 (no class)				
7	13	29 Sep 2026	Tue	Sections 5.1 & 5.2
	14	2 Oct 2026	Fri	Sections 5.2 & 5.3
8	15	6 Oct 2026	Tue	Sections 5.4 & 5.5 *(Fri 9 Oct: Well-Being Day, no class)*
Week 9 — Tue 13 & Fri 16 Oct 2026 (no class)				

Week	*Lesson*	*Date*	*Day*	*Sections covered*
10	16	20 Oct 2026	Tue	Section 6.1
	17	23 Oct 2026	Fri	Sections 6.2 & 6.3
11	18	27 Oct 2026	Tue	Sections 6.4 & 6.5
	19	30 Oct 2026	Fri	Section 7.1
12	20	3 Nov 2026	Tue	Sections 7.1 & 7.2
	21	6 Nov 2026	Fri	Revision: Chapters 1 & 2
13	22	10 Nov 2026	Tue	Revision: Chapters 3 & 4
	23	13 Nov 2026	Fri	Revision: Chapters 5–6

Email address: ch_lo@nus.edu.sg